OPEN STANDARDS

Call Detail Record (CDR)

Field Definitions & API Guidelines

Version 1.0.0 | Companion to cdr-schema.yaml (OpenAPI 3.1)

Designed for interoperability between telephony/UC switch vendors and compliance recording platforms

1. Purpose & Scope

This document defines the canonical field set for Call Detail Records (CDRs) as part of an open interoperability standard. It is intended for switch vendors, UC platform developers, and compliance recording integrators.

The schema is designed to satisfy the data requirements of compliance recording and analytics platforms while remaining broadly applicable to any recording, quality management, or analytics consumer.

Companion artefacts:

- cdr-schema.yaml — OpenAPI 3.1 machine-readable schema
- Interactive Schema Explorer — browser-based field reference

2. Field Status Key

Field Name	JSON Key	Status	Type	Description	Example
Required				Field MUST be present in every response. Absence constitutes a schema violation.	
Optional				Field SHOULD be included where the platform has the data. Omission is acceptable.	
Derived				Field is computed from other fields (e.g. duration). Provided for consumer convenience.	

3. CallRecord — Root Object

The CallRecord object represents a single interaction leg or session. For multi-party scenarios (conferences, transfers, barge-in, supervision) all parties are captured within the participants array.

3.1 Identity Fields

Field Name	JSON Key	Status	Type	Description	Example
Call ID	callId	Required	string (UUID)	Globally unique, stable call identifier. Must match the Call ID delivered to the voice recorder via CTI. Must not be recycled over time. Uniqueness must be guaranteed at minimum within a Tenant.	cc2e4b91-3f7a-4d2e-b123-8f0a1c9e5d73
Parent Call ID	parentCallId	Optional	string (UUID)	Call ID of the parent leg this leg was spawned from (e.g. on transfer or barge). Used to reconstruct related call chains.	a1b2c3d4-0000-0000-0000-111122223333
Related Call IDs	relatedCallIds	Optional	string[]	Additional related call IDs (e.g. conference legs, consultation legs).	["leg-001","leg-002"]
Tenant ID	tenantId	Optional	string	Tenant / partition identifier for multi-tenant deployments.	tenant-london

3.2 Timing Fields

All datetime values MUST be expressed in UTC using ISO 8601 format: YYYY-MM-DDTHH:MM:SS.sssZ

Field Name	JSON Key	Status	Type	Description	Example
Interaction Start Time	interactionStartTime	Optional	datetime (UTC ISO 8601)	When the interaction was initiated (dial/ringing). May be omitted if always identical to callStartTime.	2024-06-01T08:00:00.000Z
Call Start Time	callStartTime	Required	datetime (UTC ISO 8601)	When the call was connected / answered. This is the primary timing anchor for reconciliation.	2024-06-01T08:00:05.000Z

Field Name	JSON Key	Status	Type	Description	Example
Call End Time	callEndTime	Required	datetime (UTC ISO 8601)	When the call ended. Absence implies callState=ongoing.	2024-06-01T08:12:47.000Z
Interaction End Time	interactionEndTime	Optional	datetime (UTC ISO 8601)	When the full interaction ended (e.g. after wrap-up). May be omitted if identical to callEndTime.	2024-06-01T08:13:10.000Z
Last Update Time	lastUpdateTime	Optional	datetime (UTC ISO 8601)	Last modification time of this record. Used for incremental polling and ordering. Preferred sort key for API responses.	2024-06-01T08:13:10.500Z
Duration (Seconds)	durationSeconds	Derived	number (float)	callEndTime - callStartTime in seconds. Provided for convenience.	762.0

3.3 State & Classification Fields

Field Name	JSON Key	Status	Type	Description	Example
Call State	callState	Required	enum	ongoing | ended | missed | abandoned. May be inferred from absence of callEndTime (= ongoing).	ended
Call Direction	callDirection	Optional	enum	inbound | outbound | internal | unknown.	inbound
Call Type	callType	Optional	enum	peer_to_peer | group | conference | meeting | intercom | private_wire | voicemail | unknown.	peer_to_peer
Media Type	mediaType	Required	enum	voice | video | chat | instant_message | email | unknown. Primary modality. API MUST support filtering by this field.	voice

4. Participant Object

Each CallRecord contains a participants array with one entry per party. Standard two-party calls have two entries (caller + callee). Multi-party scenarios add further entries with appropriate roles.

4.1 Identity & Role

Field Name	JSON Key	Status	Type	Description	Example
Participant ID	participantId	Required	string	Unique ID for this participant within this call record. Stable within the record.	p1
Role	role	Required	enum	caller | callee | conference_participant | transfer_source | transfer_target | transfer_consultation | barge_agent | monitor_supervisor | ivr | queue | voicemail | unknown.	caller
Extension	extension	Required	string	Dialled number / extension. ANI for caller, DNIS for callee.	2341
User ID	userId	Req (internal)	string	Unique switch-side user identifier (username, AgentId, TraderId). Omit for external parties.	jsmith
Display Name	displayName	Optional	string	Human-readable display name.	John Smith
Device ID	deviceId	Optional	string	Unique device/client ID as known to the switch.	device-0042
Group	group	Required	string	Group or Skill Group this participant belongs to. Must be filterable at the API level.	trading-floor-london
Recording Config	recordingConfig	Optional	enum	Switch-side recording configuration: record | do_not_record | not_applicable | unknown.	record

4.2 Timing (per Participant)

Field Name	JSON Key	Status	Type	Description	Example
Join Time	joinTime	Optional	datetime (UTC ISO 8601)	When this participant joined. Equals callStartTime for original caller/callee. Useful for late joiners in conferences.	2024-06-01T08:00:05.000Z
Leave Time	leaveTime	Optional	datetime (UTC ISO 8601)	When this participant left. Equals callEndTime for original caller/callee.	2024-06-01T08:12:47.000Z

4.3 Device Information (DeviceInfo sub-object)

Field Name	JSON Key	Status	Type	Description	Example
Device Model	device.model	Optional	string	Device model name (e.g. 'XX SoftPhone v4', 'ABC Turret').	XYZ UC Client
Software Version	device.softwareVersion	Optional	string	Software or firmware version string.	5.2.1.1042
MAC Address	device.macAddress	Optional	string	MAC address of the device.	00:1A:2B:3C:4D:5E
IP Address	device.ipAddress	Optional	string	IP address of the device at call time.	10.0.1.45
Audio Codec	device.audioCodec	Optional	string	Audio codec negotiated for this participant.	G.722
Video Codec	device.videoCodec	Optional	string	Video codec negotiated for this participant (if applicable).	H.264

4.4 Trading-Specific Fields

Field Name	JSON Key	Status	Type	Description	Example
Slot Number	slotNumber	Optional	string	Trading turret slot number. Trading deployments only.	slot-3
Handset Info	handsetInfo	Optional	string	Handset or speaker identifier in use. Trading deployments only.	HS1

5. CallEvent Object

The events array provides an ordered timeline of discrete state changes within an interaction. It allows compliance systems to reconstruct the full call flow including transfers, conferences, recording actions, and supervisory events.

Field Name	JSON Key	Status	Type	Description	Example
Event Time	eventTime	Required	datetime (UTC ISO 8601)	UTC timestamp of the event.	2024-06-01T08:03:15.000Z
Event Type	eventType	Required	enum	See Section 5.1 for full enum values.	hold
Participant ID	participantId	Optional	string	ID of the participant who triggered this event.	p1
Target Participant ID	targetParticipantId	Optional	string	ID of the participant targeted by this event.	p2
Detail	detail	Optional	string	Human-readable detail string (e.g. IVR menu, function key, audio device name).	Transfer to extension 3301
Metadata	metadata	Optional	object (key-value)	Additional event-specific key-value pairs.	{"queue":"ACD-FX"}

5.1 Event Type Enum Values

The following eventType values are defined:

Field Name	JSON Key	Status	Type	Description	Example
ringing				Call is ringing at the destination.	
connected				Call was answered / connected.	
disconnected				Call was terminated / hung up.	
missed				Call rang but was not answered.	
hold / resume				Call placed on or resumed from hold.	

Field Name	JSON Key	Status	Type	Description	Example
transfer_initiated / completed / failed				Transfer lifecycle events.	
conference_created				A conference bridge was created from this call.	
participant_join / leave / rejoin				Participant joined, left, or rejoined a multi-party call.	
barge_in				A supervisor or agent barged into the call.	
monitor_start / stop				Silent monitoring of the call started or stopped.	
swap_audio_device				Audio device was swapped (e.g. HS1 to Speaker).	
record_on_demand_start / stop				Manual record-on-demand initiated or stopped.	
pause_recording / resume_recording				Recording paused (e.g. for PCI compliance) or resumed.	
voicemail				Call was routed to voicemail.	
park				Call was parked.	
ivr_entry / ivr_exit				Call entered or exited an IVR/IVA.	
queue_entry / queue_exit				Call entered or exited a call queue / ACD.	
pre_agent_event				Generic pre-agent event (use detail field for specifics).	
function_key_press				Client-side function key pressed (e.g. Record on Demand).	
other				Vendor-specific event not covered above. Use detail/metadata.	

6. Supplementary Objects

6.1 CallSource — Routing Origin

Provides inbound routing context for calls arriving via IVR, hunt group, or ACD queue.

Field Name	JSON Key	Status	Type	Description	Example
Hunt Number	callSource.huntNumber	Optional	string	Hunt group or original DNIS if call was redirected.	8000
IVR Info	callSource.ivrInfo	Optional	string	IVR/IVA identifier or path taken through the IVR.	ivr-main > sales
Queue Info	callSource.queueInfo	Optional	string	Call queue or ACD group the call was routed through.	ACD-FX-London
Time in IVR (s)	callSource.timeInIvrSeconds	Optional	number	Seconds spent in IVR before agent pickup.	45.0
Time in Queue (s)	callSource.timeInQueueSeconds	Optional	number	Seconds spent in call queue before agent pickup.	120.5

6.2 WrapUpInfo — Post-Call Data

Field Name	JSON Key	Status	Type	Description	Example
Wrap-Up Code	wrapUpInfo.wrapUpCode	Optional	string	Disposition / wrap-up code assigned after the call.	SALE_COMPLETED
Wrap-Up Notes	wrapUpInfo.wrapUpNotes	Optional	string	Free-text agent notes captured during wrap-up.	Customer escalated to manager.
Wrap-Up Duration (s)	wrapUpInfo.wrapUpDurationSeconds	Optional	number	Seconds spent in wrap-up state after call ended.	90.0

6.3 CloudRecordingInfo — Cloud Recording Reference

Field Name	JSON Key	Status	Type	Description	Example
Recording Status	cloudRecording.recordingStatus	Optional	enum	recorded | not_recorded | partial | unknown.	recorded
Recording Method	cloudRecording.recordingMethod	Optional	enum	automatic | manual | on_demand | unknown.	automatic
Recording ID	cloudRecording.recordingId	Optional	string	Unique recording ID as held by the cloud provider.	rec-9f8e7d6c
Media Name	cloudRecording.mediaName	Optional	string	Recording media file name.	call_20240601_jsmith.mp4
Download Path	cloudRecording.downloadPath	Optional	URI	URI to download or stream the recording.	https://recordings.example.com/rec-9f8e7d6c

6.4 QoSInfo — Quality of Service Metrics

Field Name	JSON Key	Status	Type	Description	Example
MOS Score	qos.mosScore	Optional	number (1.0–5.0)	Mean Opinion Score for the call. Higher is better.	4.2
Latency (ms)	qos.latencyMs	Optional	number	Average network latency in milliseconds.	18.5
Jitter (ms)	qos.jitterMs	Optional	number	Average jitter in milliseconds.	3.1
Packet Loss (%)	qos.packetLossPercent	Optional	number (0–100)	Packet loss as a percentage of total packets.	0.2
Total Packets	qos.packetsTotal	Optional	integer	Total RTP packets transmitted in the session.	38400

7. API Endpoint Requirements

7.1 Filtering

All CDR query endpoints MUST support the following filter parameters:

Field Name	JSON Key	Status	Type	Description	Example
startTime	query param	Required	datetime	Inclusive window start (UTC ISO 8601). Minimum granularity: per-minute.	2024-06-01T08:00:00Z
endTime	query param	Required	datetime	Exclusive window end. Records with callEndTime inside window are included even if callStartTime precedes startTime.	2024-06-01T09:00:00Z
mediaType	query param	Optional	enum (comma-list)	Filter by one or more media types.	voice,video
groups	query param	Optional	string (comma-list)	Include only records for the specified logical groups.	trading-floor-london
excludeGroups	query param	Optional	string (comma-list)	Exclude records for the specified logical groups.	non-recorded-group
X-Tenant-Id	header	Optional	string	Tenant identifier for multi-tenant deployments.	tenant-london

7.2 Ordering

Records MUST be returned in consistent ascending order. Preferred sort key is lastUpdateTime, falling back to callEndTime, then callStartTime. This enables reliable incremental polling.

7.3 Pagination

All list endpoints MUST support pagination. The response envelope MUST include:

Field Name	JSON Key	Status	Type	Description	Example
page	pagination.page	Required	integer	Current page number (1-based).	1
pageSize	pagination.pageSize	Required	integer	Number of records per page. Configurable, max 1000.	100
totalPages	pagination.totalPages	Required	integer	Total page count for this query.	4
totalRecords	pagination.totalRecords	Required	integer	Total matching record count.	382

7.4 Versioning

The API MUST publish a version string (e.g. v1, v2) as part of the URL path. In-life changes to request/response fields constitute breaking changes and MUST be published as a new major API version. Legacy versions MUST remain supported for a defined deprecation window.

7.5 Error Responses

All error states MUST be returned as structured JSON with at minimum a machine-readable code and human-readable message:

Field Name	JSON Key	Status	Type	Description	Example
code	ErrorResponse.code	Required	string	Machine-readable error code (e.g. INVALID_TIME_RANGE, AUTH_FAILED).	INVALID_TIME_RANGE
message	ErrorResponse.message	Required	string	Human-readable description of the error.	endTime must be after startTime
details	ErrorResponse.details	Optional	string[]	Optional array of additional error detail strings.	["Field: endTime"]

8. Multi-Party Call Scenarios

The CDR schema natively supports complex multi-party scenarios via the participants array, the events timeline, and the parentCallId / relatedCallIds fields. The table below summarises how each scenario is represented.

Field Name	JSON Key	Status	Type	Description	Example
Scenario	Key Fields			Representation	
Conference	participants[], events[]			Each party appears as a conference_participant with joinTime/leaveTime. participant_join and participant_leave events are emitted. relatedCallIds links all legs.	
Blind Transfer	parentCallId, events[]			Original leg has transfer_source participant and transfer_initiated / transfer_completed events. New leg references parentCallId of the original.	
Warm/Consult Transfer	parentCallId, participants[]			Consultation leg created with transfer_consultation participant. On completion the consultation leg's parentCallId points to the original call.	
Barge-In	participants[], events[]			Barge agent added as barge_agent participant. barge_in event emitted with participantId of the barge agent.	
Silent Monitor	participants[], events[]			Supervisor added as monitor_supervisor participant. monitor_start / monitor_stop events emitted. Supervisor's joinTime/leaveTime recorded.	
IVR / ACD Routing	callSource, events[]			ivr_entry / ivr_exit and queue_entry / queue_exit events emitted. callSource contains timeInIvrSeconds and timeInQueueSeconds.	
Record on Demand	events[]			record_on_demand_start / stop events emitted with participantId of the triggering agent. function_key_press event may also be emitted.	

9. Vendor Specific Fields

Vendor-specific fields not covered by this schema MAY be added to the vendorSpecificFields object at the CallRecord level, or to the metadata object within any CallEvent. Keys MUST use reverse-DNS notation to avoid collisions:

```
"vendorSpecificFields": { "com.vendorname.customField": "value" }
```

Vendor specific fields MUST NOT duplicate or override any required or optional field defined in this schema. Consumers are not required to process vendor specific fields.

10. Revision History

Field Name	JSON Key	Status	Type	Description	Example
1.0.0	2024-06-01			Initial release. Full multi-party support, QoS, cloud recording, events timeline.	